

MCT
MANJARA CHARITABLE TRUST
RAJIV GANDHI INSTITUTE OF TECHNOLOGY, MUMBAI



CODEASTRA^{2.0}

GO

BEYOND



Problem Statements

AIML • Web Dev • Cybersecurity & Blockchain • Cloud Computing

Domain : AIML

1. Crop Disease Detector from Leaf Images.

Upload a photo of a plant leaf and get a probable disease diagnosis with remedies.

2. Personal Finance Advisor Chatbot.

A conversational assistant that analyzes spending patterns from uploaded bank statements/CSV and gives saving tips.

3. Traffic Congestion Predictor.

Predicts congestion on common city routes using historical + live data for better commute planning.

Domain : Cloud computing & Distributed System

1. Intelligent Auto-Scaling for AI Workloads.

Cloud AI workloads can fluctuate dramatically. Running excessive GPU infrastructure wastes resources, while insufficient resources cause unacceptable latency.

2. Distributed Data Synchronization System.

Design a system that keeps data synchronized across geographically distributed services while handling network delays, conflicting updates and temporary disconnections.

3. Autonomous Cloud Operations Platform.

Design an intelligent cloud operations platform that continuously observes infrastructure, detects anomalies, predicts failures and recommends or executes corrective actions.

Domain : Web & Product Development

1. Blood Donor–Recipient Matching Web Platform.

A web portal connecting blood donors with recipients/hospitals in real time based on blood group and location.

2. Health Outbreak Detection & Alert System.

Analyse real-time health and public-health data to detect unusual disease patterns early and issue timely alerts about potential outbreaks.

3. Event Co-Planning Website for Friend Groups.

A website where a group can propose event ideas, vote, split costs, and finalize plans without endless group-chat back-and-forth.

Domain : Cybersecurity & Blockchain

1. Phishing Email/URL Detector.

A tool/browser extension that analyzes emails or links and flags likely phishing attempts with reasoning.

2. QR Code Safety Scanner.

Scans QR codes before opening them and warns users if the destination looks malicious.

3. Secure Voting System for College Elections.

A tamper-evident digital voting prototype for student council elections with audit logging.